

CSc 4710/6710: Assignment 2 Answer Key – Spring 2025

Problem 1 [15 = 5×3]

$F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D, D \rightarrow E\}$

1. $A^+ = \{A, B, C, D, E\}$
2. $(A, C)^+ = \{A, B, C, D, E\}$
3. Candidate Key = $\{A\}$

Problem 2 [10 = 5×2]

$F = \{PQ \rightarrow R, R \rightarrow S, S \rightarrow Q\}$

1. Candidate Key = $\{PQ, PS\}$
2. Not in BCNF ($R \rightarrow S$ and $S \rightarrow Q$ violate BCNF)

Problem 3 [10 = 5×2]

$F = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

1. Yes, the relation is in 2NF (A is single-attribute candidate key)
2. Not in 3NF ($B \rightarrow C$ violates it); Decompose into:

Option 1

- $R_1(A, B, D)$ with $A \rightarrow B, A \rightarrow D$
- $R_2(B, C)$ with $B \rightarrow C$

Option 2

- $R_1(A, B)$ with $A \rightarrow B$
- $R_2(B, C)$ with $B \rightarrow C$
- $R_3(A, D)$ with $A \rightarrow D$

xProblem 4 [20 = 5×4]

Schema: Student(StudentID, Name, Major), Course(CourseID, Title, Credits),
Enrollment(StudentID, CourseID, Grade)

- a. Students enrolled in at least 3 different courses:

```
SELECT S.Name
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
GROUP BY S.StudentID, S.Name
HAVING COUNT(DISTINCT E.CourseID) >= 3;
```

- b. Students who have taken all 3-credit courses:

```
SELECT S.Name, S.Major
FROM Student S
WHERE NOT EXISTS (
  SELECT C.CourseID
  FROM Course C
  WHERE C.Credits = 3
  AND C.CourseID NOT IN (
    SELECT E.CourseID
    FROM Enrollment E
    WHERE E.StudentID = S.StudentID
  )
);
```

- c. Student(s) with highest average grade:

```
SELECT S.Name
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
GROUP BY S.StudentID, S.Name
HAVING AVG(E.Grade) >= ALL (
  SELECT AVG(Grade)
  FROM Enrollment
  GROUP BY StudentID
);
```

- d. Students who never took any course with fewer than 3 credits:

```
SELECT S.Name
FROM Student S
WHERE NOT EXISTS (
  SELECT 1
  FROM Enrollment E
  JOIN Course C ON E.CourseID = C.CourseID
  WHERE E.StudentID = S.StudentID AND C.Credits < 3
);
```

Problem 5 [15 = 5×3]

Schema: Movie(MovieID, Title, Genre, Rating), Review(ReviewID, MovieID, ReviewerName, Score)

- a. Titles with higher avg score than every 'Horror' movie:

```
SELECT M.Title
FROM Movie M
JOIN Review R ON M.MovieID = R.MovieID
GROUP BY M.MovieID, M.Title
HAVING AVG(R.Score) > ALL (
  SELECT AVG(R2.Score)
  FROM Movie M2
  JOIN Review R2 ON M2.MovieID = R2.MovieID
  WHERE M2.Genre = 'Horror'
  GROUP BY M2.MovieID
);
```

- b. Movies never reviewed:

```
SELECT Title
FROM Movie
WHERE MovieID NOT IN (
  SELECT DISTINCT MovieID
  FROM Review
);
```

- c. Reviewers who reviewed every movie:

```
SELECT R1.ReviewerName
FROM Review R1
```

```
GROUP BY R1.ReviewerName
HAVING COUNT(DISTINCT R1.MovieID) = (
    SELECT COUNT(DISTINCT MovieID)
    FROM Movie
);
```